

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		See example graphs at end of Q6 Award 3 marks for all points and error bars correct Award 2 marks for 5 points and error bars correct Award 1 mark for 4 points and error bars correct Award 0 marks for 3 points or less and error bars correct Correct lines drawn for the maximum gradient and the minimum gradient (1) ecf		4		4	4	4
		(iii)		l can be measured to ± 0.001 [m] or resolution is 1 mm (1) [Uncertainty in $\sqrt{l}$] is too small to plot (1)			2	2		2
		(iv)		Both gradients correct (1) Mean gradient correct (1) ecf Absolute uncertainty in gradient correct (1) ecf implied by % uncertainty Correct g (1) ecf Final % correct (1) ecf No sig fig or unit penalties $m_{\max} \left[= \frac{58.2}{1.40 - 0.06} \right] = 43.4; m_{\min} \left[= \frac{54.3 - 4.7}{1.4} \right] = 35.4$ $\therefore \text{Gradient, } m = 39.4 \pm \{4.0 / 10\}\%$ Either For 20 oscillations, $T = 20 \times 2\pi \sqrt{\frac{l}{g}}$ $\therefore \sqrt{g} = \frac{40\pi}{m} \text{ [or by impl.] [= } 3.19 \pm 10\%]$ $\therefore g = 10.2 \text{ m s}^{-2} \pm 20\%$		5		5	5	5